

Examiner reports

Q1.

Here most correctly identified the end points of the interval. Many included the end points in their interval and so lost the accuracy mark.

Q3.

This final question resulted in a wide spread of marks. In part (a) a significant number of students did not see that the answers could be written down directly. The most common incorrect method led to incorrect equations similar to $x + 30 = 89.27$, the right-hand side coming from $\tan^{-1}(79)$. Most students correctly recognised the geometrical transformation in part (b) to be a translation but the direction of the translation was quite frequently incorrect. In part (c)(i) the right-hand side of the given equation was usually expanded correctly with the correct notation seen and many recalled and used the correct relevant trigonometric identity to eliminate $\sin^2 \theta$. However, solutions towards the printed answer were spoilt in a significant minority of cases by students incorrectly writing the correct expression $5 + 1 - \cos^2 \theta$ as $5 - 5\cos^2 \theta$. Of those who obtained the correct simplified quadratic equation in $\cos \theta$, most were able to solve the equation correctly but not all of these students gave an indication that they had considered and rejected the value -2 for $\cos \theta$. The final part of this last question proved to be a challenge even for the more able

students. Many solutions just consisted of solving $\cos \theta = \frac{3}{4}$ to obtain two solutions rather than stating the required equation as $\cos 2x = \frac{3}{4}$ and solving it to find the four solutions within the given range to the required degree of accuracy.

Q4.

Sketching a convincing tangent graph, even with graphical calculators available, seemed a considerable challenge for many. Three separate branches were needed with the correct intercepts on the x -axis and demonstration of the tendency near 90° and 270° . It was disappointing to sometimes see a sine or a cosine graph drawn instead. Many students correctly solved the trigonometric equation in part (a)(ii) although a minority of students incorrectly gave more than two solutions inside the given interval. The majority of students were able to recall and use the required two trigonometric identities in part (b)(i) but not all were able to reach the printed answer without there being a sign error in their working. In the final part of this last question on the paper it was important that students recognised that they were required to use 'Hence'. and also that the equation to be solved was in terms of $3x$. Those students who just solved the equation in part (b)(i) without ever dividing its solutions by 3 or writing ' $\theta = 3x$ ' were heavily penalised. Those students who did link θ to $3x$ generally recognised and solved the quadratic in $\cos 3x$ but a significant minority then failed to find a third solution for x in the given interval.

Q5.

The response to this question was very mixed, due to a student's understanding, or misunderstanding, of the term exact. Despite $\tan \alpha$ being given as a surd in the question, many students interpreted *exact* as meaning a number with many decimal places. As a result, many students, nearly 30% of the entry, scored at most 2 of the 8 marks available,

only gaining credit for showing some knowledge of the required trigonometric formulae and nothing else. A similar proportion, however, had the question fully correct.

- (a) (i) Angle α was given as acute and it was expected students would use a right angle triangle to show the requested value of $\sin \alpha$. Many did this, but many found the angle itself using inverse tangent and then its sine for no credit;
- $(\tan^{-1} \frac{2}{\sqrt{5}}) = \frac{2}{3}$
- similarly $\sin$ gained no credit. Many found $\cos \alpha$ from $\sin \alpha = \cos \alpha \tan \alpha$ which was accepted, if exact. Some students successfully derived the result using $\operatorname{cosec}^2 \alpha = 1 + \cot^2 \alpha$.
- (ii) The vast majority of students knew, or could derive, the double angle formula for $\sin 2\alpha$, and so obtained the correct exact value if they had $\cos \alpha$ correct.
- (b) Most students knew the required compound angle formula, although occasionally there was a sign error and some thought the formula was $\cos \alpha - \cos \beta$. Many approaches were used to find $\sin \beta$ and $\cos \beta$. Some used the simplest method, involving a right angle triangle, whilst some noticed that $\sin \beta / \cos \beta = \frac{1}{2}$ and used this in conjunction with $\sin^2 \beta + \cos^2 \beta = 1$ to find the values. Most students who had the four component parts correct could complete to the requested form; but some were unable to manipulate the surds as required, even from $\frac{2}{3} + \frac{2}{15} \sqrt{5}$. Many students who had worked with inexact values, wrote down $k = 5$ as their final answer, but such students need to understand that using results from their calculator does not show this, as requested in the question.

Q6.

Most candidates had part (a)(i) correct, although the amount of work they did varied considerably. Some just wrote down the value of $\tan \alpha$, presumably recognising the 3-4-5 triangle, whereas others used Pythagoras. Several weaker candidates divided $\sin \beta$ by

$\cos \alpha$ to give an answer of $\frac{5}{6}$. Some candidates used trigonometric identities, some finding $\sin \alpha$ first and using it together with the given cosine to calculate the tangent.

In part (a)(ii), the majority of candidates failed to work with a negative value for $\tan \beta$, despite most of them showing awareness of an obtuse angle through quoting 150° . These candidates then could not get to the right answer in part (b).

However, in part (b), most candidates did use the compound angle formula for $\tan (\alpha + \beta)$ correctly, with only a few sign errors seen and some leaving the $\pm$, from the formula book, in their expression. Most tried to manipulate the surds in their expression towards the required form of the answer, but with varying degrees of success. Those who inverted the denominator and then multiplied the resulting fractions were the more successful. Some who had worked with the correct value of $\tan \beta$ failed to get to the required form by not seeing that they could cancel or multiply through their expression by a factor of $\sqrt{3}$.

Q7.

This question, which tested the basic trigonometry section of the specification, gave the value of $\sin \theta$, enabling the area of the triangle in part (a) to be evaluated simply and exactly. This was misinterpreted as a value of the angle θ by some students, for which generous allowance was made in this and the final part. Part (b) specifically asked for the exact value of $\cos \theta$, the significance of which was ignored by many. Those who used the identity $\cos^2 \theta = 1 - \sin^2 \theta$ generally obtained the correct value and full credit was also given for a correct exact value straight from the calculator. The cosine formula was generally well-known and correctly written down in part (c), but there was a substantial minority of students who used the wrong order of evaluation, which resulted in $AC^2 = 30.25 \cos \theta$. A higher proportion of students than usual seemed to have their calculators set in the wrong mode. The final mark was reserved for those calculating the length of AC without unnecessary approximation.

Q8.

In part (a)(i), most candidates stated the formula for $\sin 2\theta$ correctly although not always in its simplest form.

In part (a)(ii), although many candidates knew the formula for $\cos 2\theta$ in at least one its correct forms, several gave $\cos^2 \theta + \sin^2 \theta$, and later in the question just gave the value of $\cos 2\theta$ as 1 without apparently thinking something could be wrong.

In part (a)(ii), many candidates had problems. It appeared that the 3-4-5 triangle was either not well known or not recognised as relevant; although some candidates did draw this to justify their answer others used $\sin^2 \theta = 1 - \cos^2 \theta$. Many candidates worked backwards from the answer given, to find a value for $\sin \theta$, which gained no credit nor did going via an angle from the given value of $\cos \theta$. Many candidates omitted to find $\cos 2\theta$ or, as above, gave the value as 1, or another incorrect value from an incorrect expression. Those who did find the correct value for $\cos 2\theta$ often dropped the minus sign in later working.

In part (b)(i), the use of the chain rule with these parametric equations was generally done well with most candidates completing it correctly. Some candidates apparently tried to integrate rather than differentiate and some made sign errors when differentiating. A few candidates had the chain rule expression the wrong way up.

In part (b)(ii), although some neat and concise correct answers were seen, many candidates got in a mess trying to use angles and not working in exact fractions. Some confused the value of the sine and cosines involved with the angles writing terms such as $\sin \frac{3}{5}$. Some left a mix of fractions and decimal numbers in their final answer, which is bad practice but was accepted if the values of the coordinates at P were correct.

Q9.

This question, which tested trigonometric-graphs, identities and equations, was better answered than similar questions in some recent series. In part (a) most of the graphs presented had the correct shape with intercepts given, although the intercept $y=1$ was occasionally omitted and those on the x -axis only expressed in degrees. Part (b)(i) usually

had enough to justify the given result which was then generally used in part (b)(ii). The most common error in the final part was to omit the second answer for x . It was not uncommon to see weaker candidates writing 'identities' such as ' $\cos 2x \times \cos 2x = \cos^2 4x$ '.

Q10.

Many candidates found the inverse sine of 0.8 correctly but a significant minority did not give the second solution. Parts (b)(i) and (b)(ii) were usually well answered but wrong signs, use of degrees and reversed coordinates were all seen in part (b)(ii). Part (b)(iii) was frequently not attempted or marks were lost due to an apparently careless omission of brackets. Sketches for $\sin 2x$ resulted in a spread of marks awarded. Those sketches gaining no credit usually resulted from stretching the given graph of $\sin x$ with a scale factor 2 parallel to the y -axis.

One mark was the usual award for those stretching the graph of $\sin x$ parallel to the x -axis but with scale factor 2. Those who correctly stretched $\sin x$ parallel to the x -axis with scale factor $\frac{1}{2}$ scored at least two marks, and more often at least four marks, with a significant number failing to get the fifth mark because the x -coordinates of the maximum points were not shown.

Q11.

It was anticipated that candidates would find most of this question accessible, but many made mistakes or showed misunderstandings of what the question parts required.

In part (a)(i) relatively few candidates could just write down the result for $\cos \alpha$; some sensibly drew a right angled triangle to help them, whilst others derived the result from $\cos^2 \alpha + \sin^2 \alpha = 1$. Many candidates, though, found the angle on their calculator and then its sine, some giving the result as 0.600.

Most candidates gave a correct expansion of $\cos(\alpha - \beta)$ for part (a)(ii) but then left that as their answer, without substituting the now known values of $\sin \alpha$ and $\cos \alpha$.

Relatively few candidates responded to the word "exact" in part (a)(iii) by using the compound angle formulae to give $\cos(\alpha - \beta)$ as a rational number. Some candidates confused the angles with the values of their sines and cosines. Other candidates just calculated angles, and gave an interpretation of "exact", as the requested cosine given to many decimal places.

In part (b)(i) most candidates used the double angle formula for tangent successfully, and went on to derive the given quadratic equation. The rarely seen errors were to confuse the denominator and numerator in the expression for $\tan 2x$ or to actually leave $\tan 2x$ in the expression in place of $\tan x$. Having derived a quadratic equation, few candidates proceeded to try to solve it by conventional methods in part (b)(ii) and some did not attempt this part of the question at all. Many manipulated the equation, either in $\tan x$ or

$\tan 22\frac{1}{2}^\circ$, in various ways before abandoning it. Some made progress through completing the square but often forgot to consider both the positive and negative square root. Of those who did solve the quadratic equation correctly, few could give a convincing

justification for rejecting the negative root, usually just writing something to the effect that the result must be positive, if they wrote anything at all.

Q12.

Most candidates were able to give a correct value in part (a) but a significant minority did not give any second value, despite the hint in the question: “answers”. A common wrong answer was “4.407”. In part (b)(i), some candidates mixed up the coordinates and in part (b)(ii) “ $(\pi + \alpha)$ ” was the usual wrong answer. The description of the required transformation in part (c) was answered better this year.

Although some excellent solutions were seen to part (d), these were very much the exception. The vast majority of candidates could not deal correctly with the $2x$ beyond

finding the first solution, $x = \frac{2\pi}{5}$.

Q13.

In part (a), the question asked for the coordinates of the marked points on the trigonometrical graph. For each point, examiners expected to be given both coordinates rather than just a non-zero value. Other common errors included answers involving radians and also answers which had the x - and y -coordinates switched. In part (b), candidates were expected to use the word “stretch” and also to give the direction and scale factor of the stretch. As expected, a scale factor of 2 rather than $1/2$ was the usual error.

Relatively few candidates gained all the available marks in part (c). Most obtained two

of the four values, but many missed the value from ‘ $x = 180 + 180 - \frac{\alpha}{2}$ ’. Candidates should also be aware that when asked to give answers to the nearest 0.1° , their earlier working should be done to a greater degree of accuracy. Failure to do this frequently lead to the wrong answer “214.2°”.

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